

Class Notes CIS 675
Artificial Intelligence: Overview

AI: Overview

- An Attempt to classify
- Foundation
- Modern topics

An Attempt to Classify

- Two approaches: human-centered vs. rationality. Both directions have yielded some valuable insights.
 - A human-centered approach is empirical science, involving hypothesis and experimental confirmation.
 - A rationalist approach involves a combination of mathematics and engineering.
- Both approaches can be broken up into fields that are concerned with
 - the thought process and reasoning, or
 - the behavior.

Four Categories of AI

Systems that think like humans.	Systems that think rationally.
Systems that act like humans.	Systems that act rationally.

Acting humanly

- **Turing Test**, proposed by Alan Turing (1950). Turing's definition of intelligent behavior as the ability to achieve *human-level performance* in all cognitive tasks.
- The test involves an interrogator who interacts with one human and one machine. Within a given time the interrogator has to find out which of the two is the human, and which one the machine.
- Issue to solve in order to pass the test using a terminal: natural language processing, knowledge representation automated reasoning, machine learning.
- Additional requirements for the “total Turing test”: computer vision, speech recognition, speech synthesis, robotics.

Thinking humanly

- Need to understand how humans think.
- Methods:
 - catch our thoughts while they go by
 - psychological experiments
- Example: GPS, “General Problem Solver” (Newell and Simon, 1961). Objective to compare the steps of the problem solving with the reasoning of human subjects solving the same problem.
- **Cognitive Science** embraces computer models from AI and experimental methods from psychology.

Thinking rationally

- Aristotle made first attempt to harness “right thinking”. Famous example: “Socrates is a man; all men are mortal; therefore Socrates is mortal.”
- Formal logic (late nineteenth, early twentieth centuries) provides a precise notation of statements of all kinds of things and relations between them.
- Programs that can find the solution to a logical problem if one exists have been around since 1965. Given enough memory and time, they should be able to solve everything.
- Problems:
 - not everything can be formalized, e.g. uncertain knowledge
 - complexity of problems easily exhausts today's (and tomorrow's) computational resources.

Acting rationally

- “... acting so as to achieve one’s goals, given one’s beliefs.”
- Objective: construction of rational **agents**.
- AI as rational agent design is more general than the “laws of thought” approach; correct inference is useful but not necessary for achieving rationality.
- In contrast to human-centered approaches, the standard of rationality is clearly defined and general.

Foundation

- Philosophy (428 B.C. - present): Plato, Socrates, Aristotle ...
- Mathematics (c. 800 - present): al-Kowarazmi: algorithms, George Boole/Boule: formal language for making logical inferences (1847), David Hilbert, Kurt Gödel, Alan Turing ...
- Psychology (1879 - present): Herman von Helmholtz, Wilhelm Wundt: scientific methods to study human vision ...
- Computer Engineering (1940 - present) Alan Turing: first operational computer, Konrad Zuse: first programmable computer ...
- Linguistics (1957 - present): B. F. Skinner: verbal behavior, Noam Chomsky ...

Modern Topics

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AAAI Conferences

- Advanced Planning Technology (ARPI)
- Artificial Intelligence and Manufacturing (SIGMAN)
- Artificial Intelligence and Molecular Biology Conference (ISMB)
- Artificial Intelligence Planning Systems Conference (AIPS)
- Florida AI Research Society Conferences (FLAIRS)
- Bar-Ilan Symposium on Foundations of Artificial Intelligence
- Innovative Applications of Artificial Intelligence Conference (IAAI)
- International Conference on Multiagent Systems (ICMAS)
- Knowledge Discovery and Data Mining Conference (KDD)
- Midwest AI and Cognitive Science Society Conference (MAICS)
- National Conference on Artificial Intelligence (AAAI)
- Artificial Intelligence and Manufacturing (SIGMAN)

- Third International Conference on Multistrategy Learning (MSL)

AI Magazine

- MiTAP for Biosecurity: A Case Study Laurie Damianos, Jay Ponte, Steve Wohlever, Florence Reeder, David Day, George Wilson, and Lynette Hirschman 23(4): Winter 2000, 13-29
- Staff Scheduling for Inbound Call and Customer Contact Centers Alex Fukunaga, Ed Hamilton, Jason Fama, David Andre, Ofer Matan, and Illah Nourbakhsh 23(4): Winter 2000, 30-40
- Information Self-Service with a Knowledge Base That Learns Stephen D. Durbin, Doug Warner, J. Neal Richter, and Zuzana Gedeon 23(4): Winter 2000, 41-50
- Training and Using Disciple Agents: A Case Study in the Military Center of Gravity Analysis Domain Gheorghe Tecuci, Mihai Boicu, Dorin Marcu, Bogdan Stanescu, Cristina Boicu, and Jerome Comello 23(4): Winter 2000, 51-68
- Applying Perceptually Driven Cognitive Mapping to Virtual Urban Environments Randall W. Hill, Jr., Changhee Han, and Michael van Lent 23(4): Winter, 2002, 69-80
- Computational Vulnerability Analysis for Information Survivability Howard Shrobe 23(4): Winter, 2002, 81-94
- Intelligent Integration of Information and Services on the Web Eleni Stroulia 23(4):

Winter, 2002, 109-112

- Consciousness Constrained Paul De Palma 23(4): Winter, 2002, 122-124